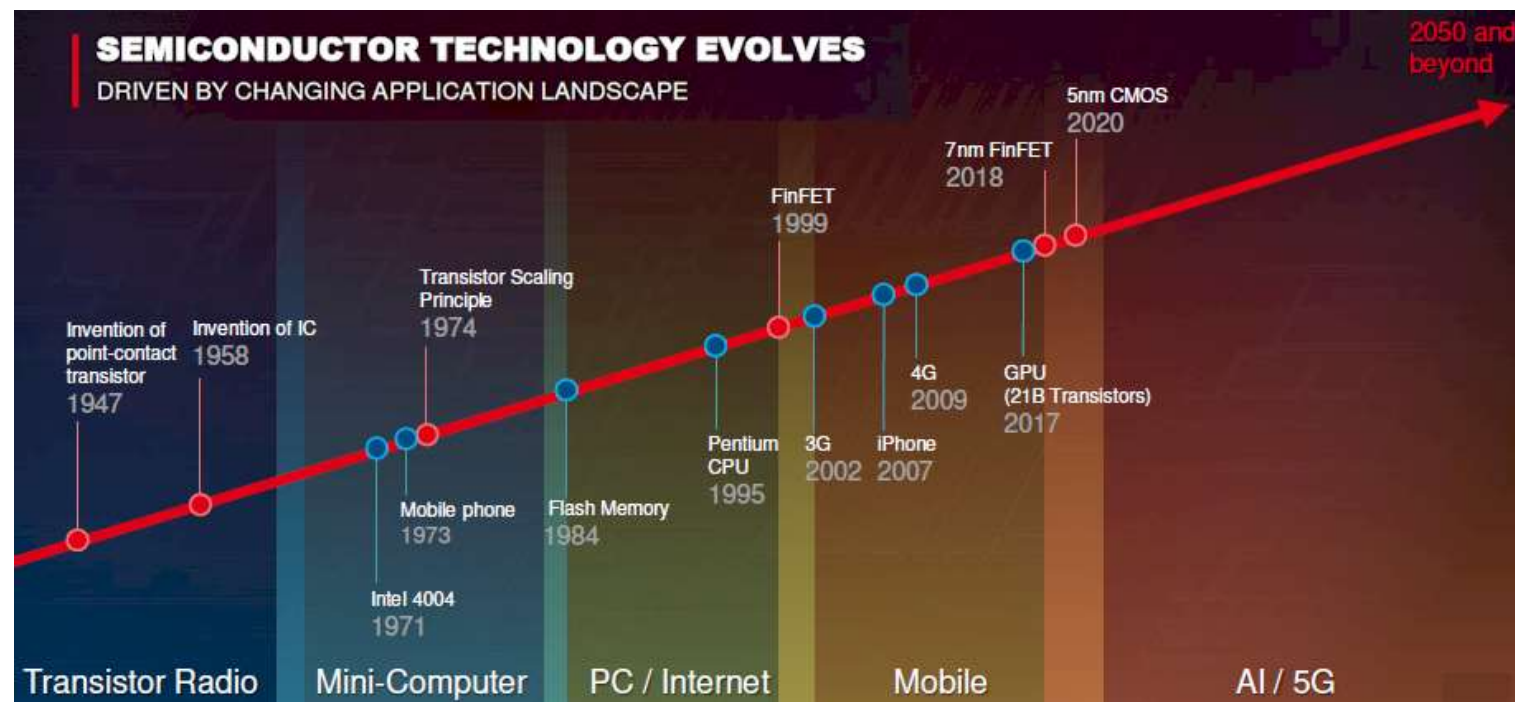


ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING

The cost of a chip is determined mainly by:

- cost of the die
(cost of manufacturing a wafer divided by the number of good dies per wafer),
- testing costs (both for die testing and testing after packaging),
- packaging costs,
- yield (percentage of functional chips).

ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING



ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING

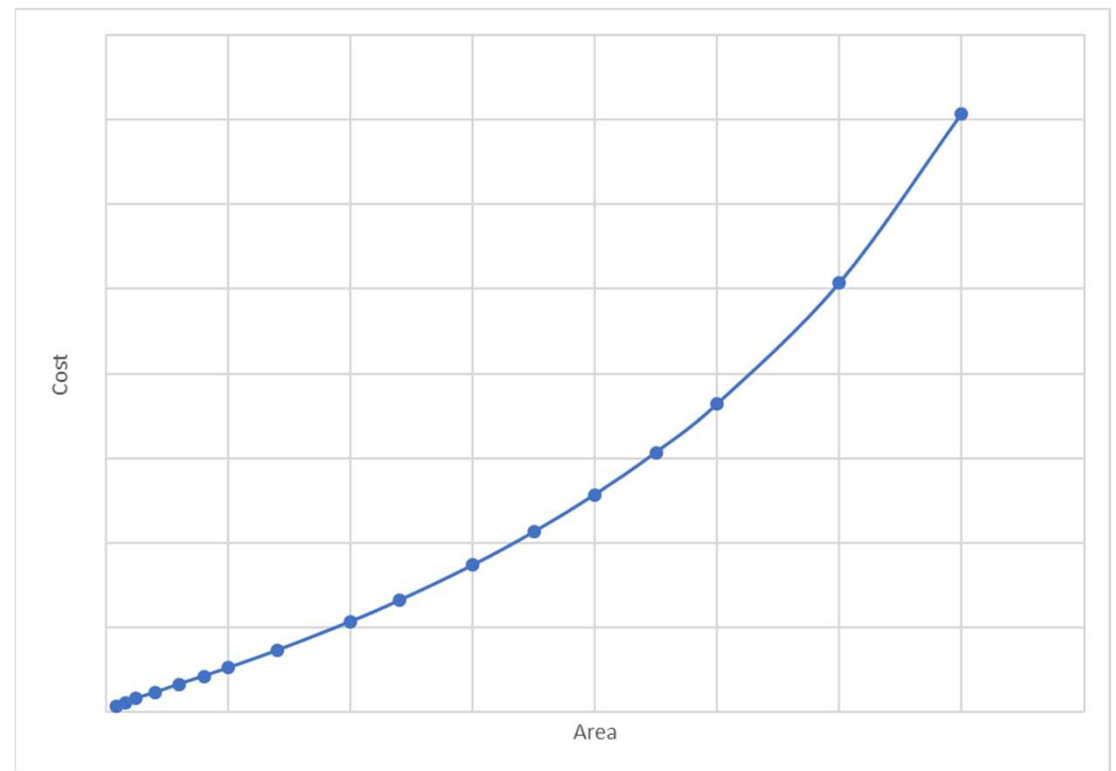
The cost per transistor reduces exponentially as the technology advances.



Image Source: Intel

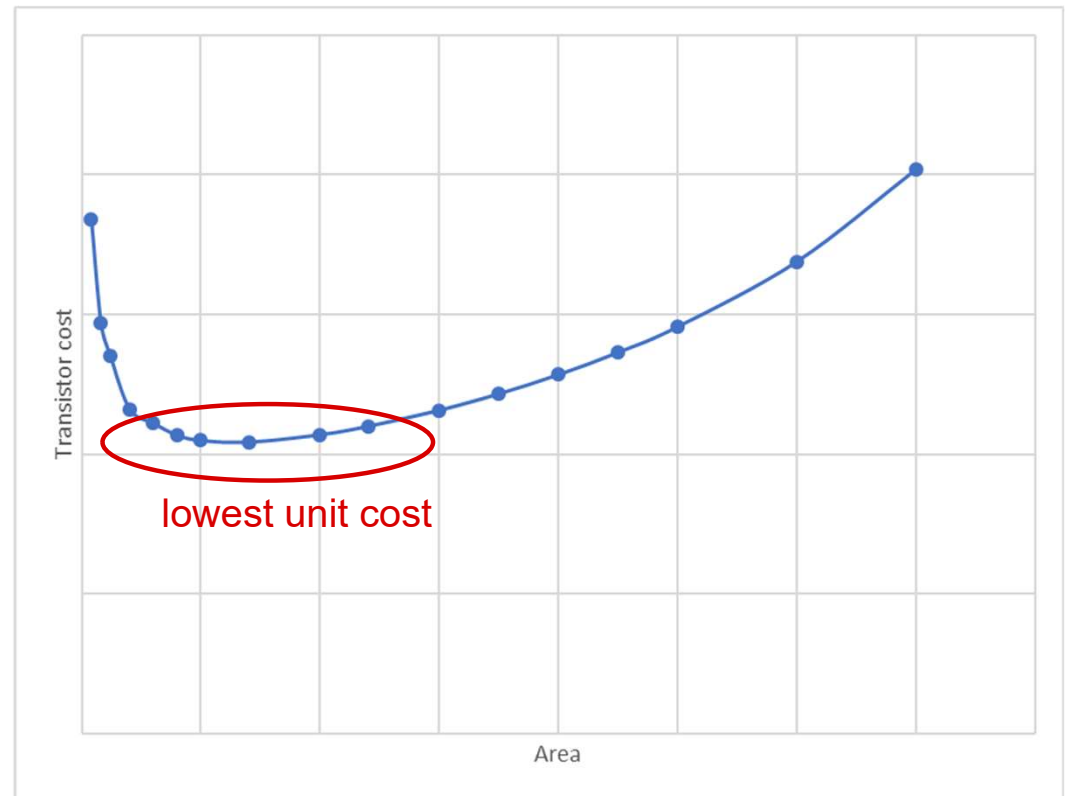
ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING

As the chip gets larger the yield reduces, hence, chip cost does not grow linearly with the area.



ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING

Plotting the cost of each transistor, as the chip gets larger, shows that there is an area range with the lower transistor costs.



ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING

Curve shows how many transistors can be purchased with one dollar for each manufacturing technology over time.

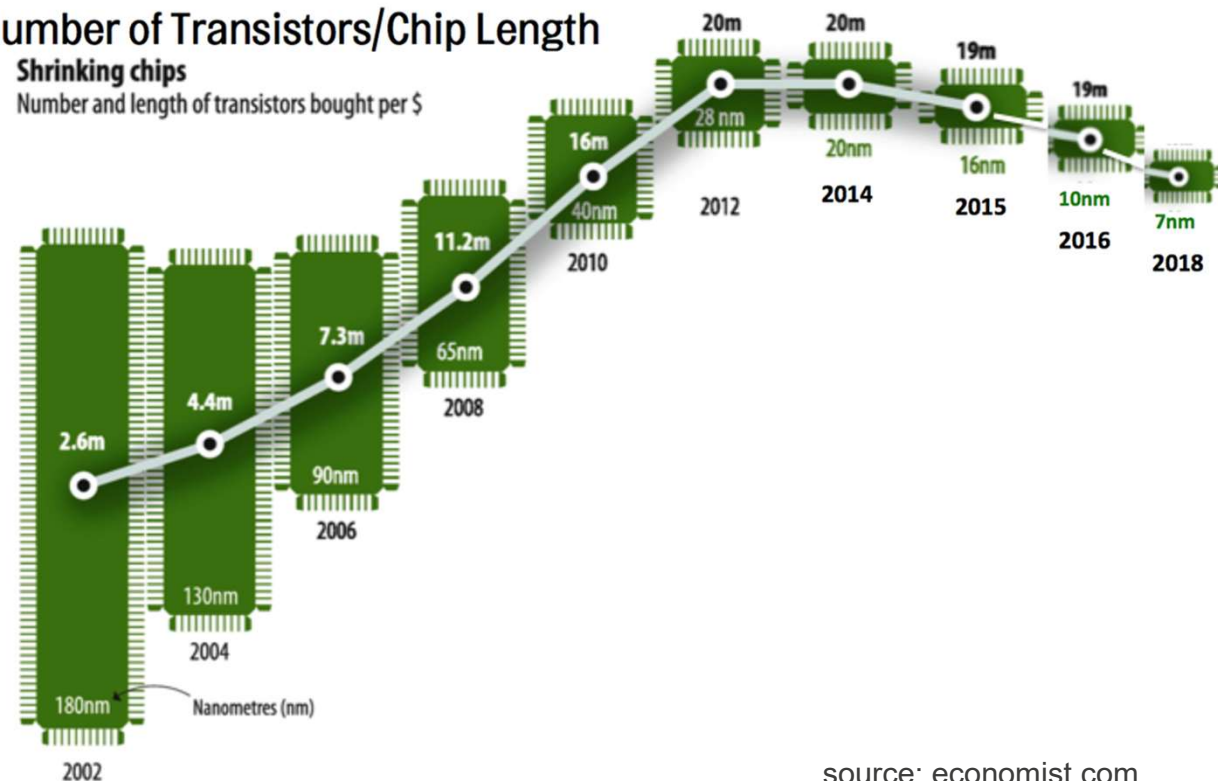
This graph represents the situation in 2018. At the time, the most cost-effective technology was 20 nm.

Over time, as the processes mature, transistors cost lower. Until a given technology enters obsolescence.

Number of Transistors/Chip Length

Shrinking chips

Number and length of transistors bought per \$



source: economist.com

ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING

In a highly competitive market, manufactures aim at the best possible selection of features:

- manufacturing process with the highest number of transistors / dollar (20 nm) per “number of transistors/chip length” curve
- chip size in the optimal region (previous slide)

These two decisions lead to a given number of transistors (this number changes over time as manufacturing processes improve).

ECONOMICS OF INTEGRATED CIRCUITS MANUFACTURING

Key design decision:

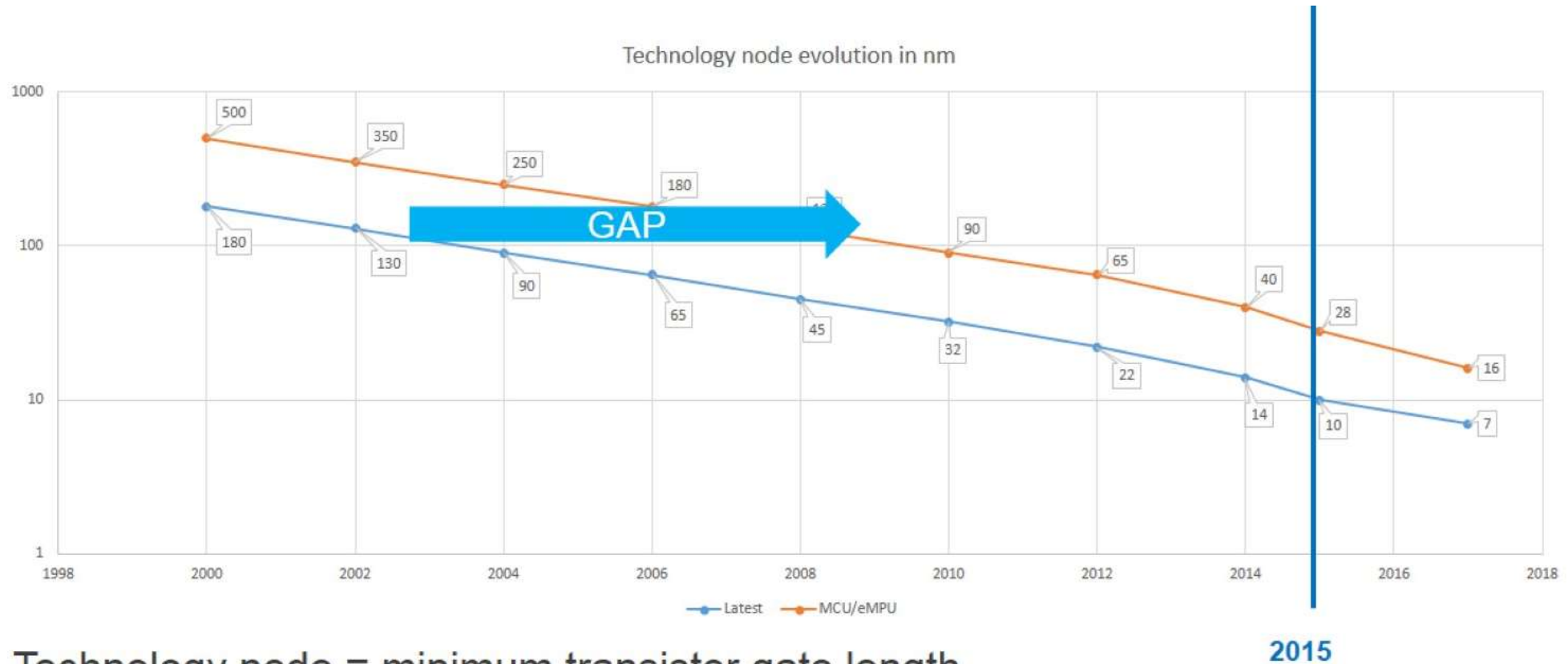
If currently the “sweet spot” (best value for money) is a given manufacturing technology and a transistor count of T millions of transistors, **what is the best use this asset ?**

1. CISC core + little Cache/Memory/Peripherals
2. RISC core + large amount of Cache/Memory/Peripherals

Most silicon vendors select the second alternative

TRENDS

Five-year gap from state of the art to MCU technology



Technology node = minimum transistor gate length

Source: ITRS, Renesas Electronics – based on product announcement

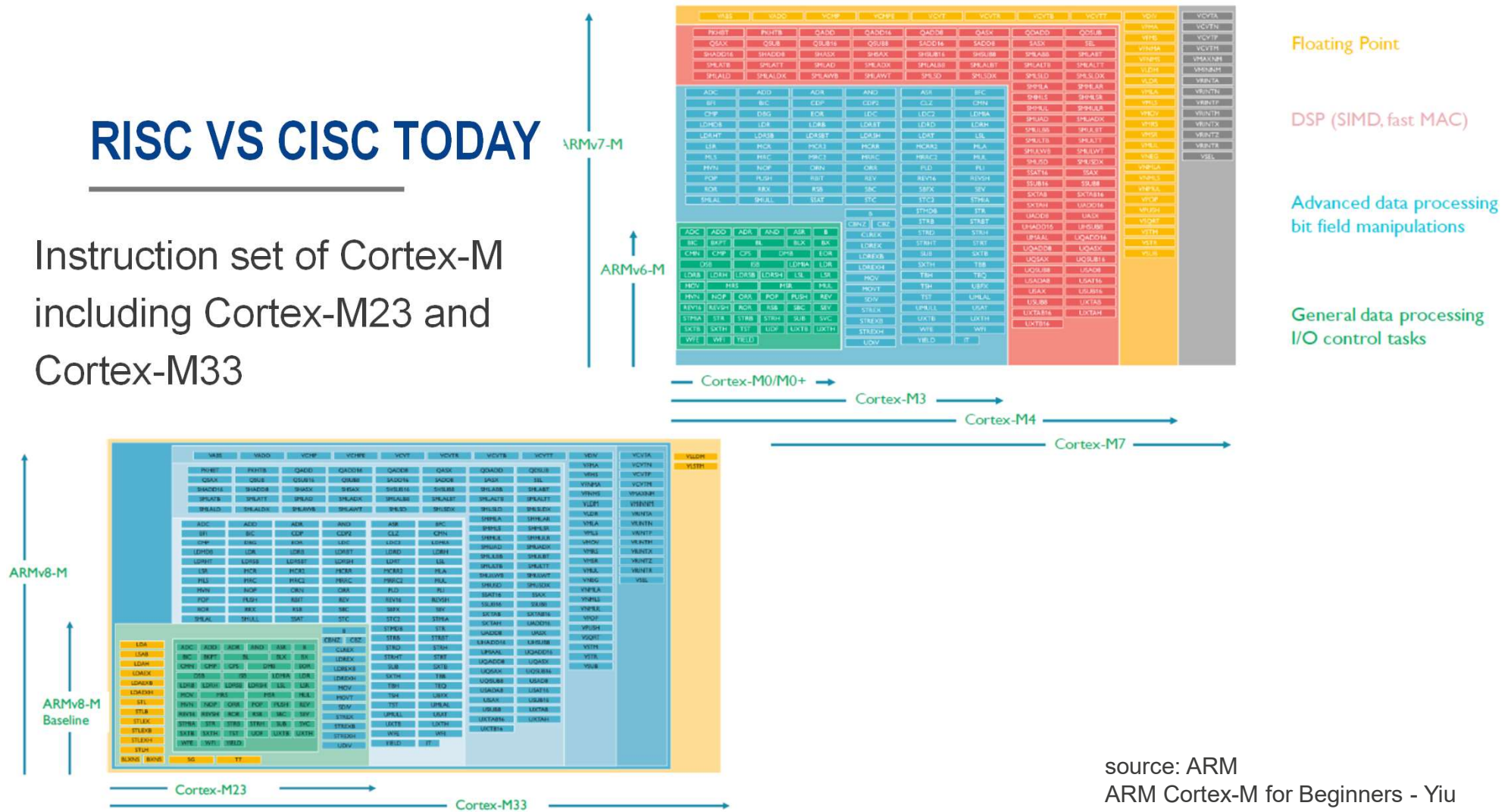
RISC VS CISC TODAY

Currently, there is a trend to mix the best features of both paradigms. The convergence of RISC and CISC has not been named yet, but the characteristics of novel processors are:

- large instruction set
- regular instructions, prone to pipelining
- pipelines from 3 to 20 stages
- no microcode

RISC VS CISC TODAY

Instruction set of Cortex-M including Cortex-M23 and Cortex-M33



source: ARM
ARM Cortex-M for Beginners - Yiu